

RECOMMENDED MATERIAL · RANK #1

High-Barrier Laminate

Multilayer laminate · Not readily recyclable · Not biodegradable · Relative cost: High

89.2

SUITABILITY SCORE / 100

High-Barrier Laminate ranks first among 6 compatible material(s) for Chips (Snacks) stored ambient at 25 °C / 65% RH, with a suitability score of 89.2/100. Main contributing criteria: food compatibility and storage compatibility. 3 material(s) were filtered out as incompatible.

Suitability score = weighted rule-based compatibility score; not a model accuracy figure.

Input Parameters

Parameter	Value	Parameter	Value
Commodity	Chips	Food category	Snacks
Moisture content	2%	Oil / fat content	35%
pH	6.5	Respiration rate	Low
Moisture sensitivity	High	Oxygen sensitivity	High
Desired shelf life	120 days	Storage	Ambient at 25 °C
Relative humidity	65%	Transportation	Long Distance
Cost priority	Balanced	Sustainability priority	Medium
MAP required	Not sure	Engine MAP mode	Gas-flush MAP

Packaging Specifications

Property	Value	Target / note
OTR	0.5 – 5 cm³/m²·day	Very low (< 5 cm³/m²·day) - Meets target
WVTR	1 – 5 g/m²·day	Very low (< 1 g/m²·day) - Partially meets
Recommended thickness	120 µm	Material range 60 – 150 µm
Sealability	Excellent (5/5)	
Gas permeability	Very low (near-hermetic) (1/5)	
Mechanical strength	Excellent (5/5)	
MAP suitability	Excellent (5/5)	
Suggested format	Hermetically heat-sealed pouch with nitrogen gas flushing (MAP suggested by engine)	

Reference ranges: OTR at 23 °C, 0% RH; WVTR at 38 °C, 90% RH (typical test conditions)

Why This Material?

- + Listed for Snacks applications in the packaging knowledge base
- + Rated for ambient conditions at 25 °C
- + Oxygen barrier (5/5) meets the required ~5.0/5
- + Protection index 4.6/5 supports the 120-day shelf-life target
- + Supports gas-flush MAP (oxygen barrier 5/5, sealability 5/5)

Score Breakdown

Criterion	Criterion score	Effective weight	Contribution
Food compatibility	100%	25.0%	25.0
Storage compatibility	100%	20.0%	20.0
Barrier requirements	94%	20.0%	18.8
Shelf-life suitability	100%	15.0%	15.0
Cost	51%	10.0%	5.1
Sustainability	20%	5.0%	1.0
MAP suitability	85%	5.0%	4.3

Key Decision Factors

Factor	Input	Derived requirement	Influence
Respiration rate	Low	Non-respiring product: hermetic barrier packaging is acceptable	Low
Moisture sensitivity	High	Moisture barrier $\geq 4.5/5$ (high sensitivity)	High
Oxygen sensitivity	High	Oxygen barrier ~5.0/5 (high sensitivity, 35% fat, risk of rancidity)	High
Storage temperature	25 °C (ambient)	Material must be rated for ambient conditions	Medium
Relative humidity	65%	Humidity within a manageable range	Low
Shelf-life target	120 days (medium)	Protection level ~3.5/5 required	Medium
Transportation	Long-Distance	Mechanical strength $\geq 4.5/5$ suggested	High
MAP requirement	Not sure	Active MAP (N ₂ /CO ₂ gas flushing) - suggested by engine	Medium
Oil/fat content	35%	Oxidation risk raises the oxygen-barrier requirement	Medium

Alternative Materials

Rank	Material	Score	Main advantage	Main trade-off
#2	Metallized Film	87.9	Higher cost score than High-Barrier Laminate (68% vs 51%)	Oxygen barrier (4/5) is below the required ~5.0/5
#3	Aluminum Foil Laminate	87.5	Oxygen barrier (5/5) meets the required ~5.0/5	Very high relative material cost

Trade-offs to Consider

- High-Barrier Laminate: High relative material cost
- High-Barrier Laminate: Multi-material structure that is difficult to recycle
- Metallized Film scores higher on cost but ranks 1.3 points lower overall
- Aluminum Foil Laminate scores higher on barrier requirements but ranks 1.7 points lower overall

Full Ranking and Filtered-out Materials

Rank	Material	Suitability score
#1	High-Barrier Laminate	89.2
#2	Metallized Film	87.9
#3	Aluminum Foil Laminate	87.5
#4	HDPE	81.8
#5	PET	80.0
#6	PP	79.5

Excluded material	Reason(s)
LDPE	Oxygen barrier (1/5) is too low for a highly oxygen-sensitive product
Biodegradable Film	Moisture barrier (1/5) is too low for a moisture-sensitive dry product
Micro-Perforated Film	Oxygen barrier (1/5) is too low for a highly oxygen-sensitive product; Perforated, breathable structure gives no barrier protection to a non-respiring product

Disclaimer

Recommendations are intended as a decision-support aid and should be validated against applicable food-contact regulations, material specifications, and laboratory/industry requirements before commercial deployment.